José R. Ocampo-Gómez

Curriculum Vitae

jocampog@stanford.edu
https://jrocampogomez.github.io/
orcid.org/0009-0009-4123-9214
https://github.com/jrocampogomez
https://www.linkedin.com/in/jrocampogomez/

RESEARCH EXPERIENCE

JUNE 2025 - PRESENT

Kavli Institute for Particle Astrophysics and Cosmology (KIPAC)

Undergraduate Researcher

AGN Reverberation Mapping:

- Advisor: Prof. Steven Allen

- Measured reverberation time lags between AGN continuum and Fe II broad-line emission across 25 OzDES-monitored active galactic nuclei ($z \approx 1-2$), to constrain the spatial extent of the Fe II emission region as a probe of supermassive black hole properties.

- Applied and cross-validated three independent time-series techniques (PyCCF, JAVELIN, & PyROA) on multi-year, irregularly-sampled photometric and spectroscopic data.

- Developed a simulation-based validation framework to quantify each method's accuracy and failure modes under realistic noise and cadence.

- Developing constraints on the Fe II radius–luminosity relation to estimate black hole masses.

AGN Luminosity Evolution:

- Advisor: Prof. Vahé Petrosian

- Determining the cosmological evolution of the gamma-ray and radio luminosity functions of flat spectrum radio quasars (FSRQs) using non-parametric methods for truncated data.

- Constructing a tri-variate sample from Fermi-LAT gamma-ray, VLBI radio, and Gaia DR2 optical observations, accounting for the additional selection bias introduced by optical redshift measurement.

- Applying the Efron–Petrosian method and Lynden-Bell C^{-} estimator to recover unbiased luminosity evolution indices and local luminosity functions.

- Measuring the intrinsic gamma-ray–radio luminosity correlation to constrain the relationship between jet and accretion-disk emission.

APRIL 2023 - APRIL 2024

NASA California Space Grant Consortium

Undergraduate Researcher

- Advisor: Prof. Jayesh Bhakta

- Built a low-cost flight telemetry and data-logging system for experimental RC aircraft, integrating inertial, barometric, GPS, and current sensors on a Raspberry Pi Pico in MicroPython.

- Implemented a packet-based nRF24L01 downlink with redundant onboard microSD logging to preserve data through radio-link loss.

- Developed a real-time ground-station display in Java/Processing, deriving aircraft pitch and roll from three-axis accelerometer data.

- Identified RF interference between the aircraft's control and telemetry transmissions as the source of downlink instability.

EDUCATION

EXP. 2027 — **Bachelor of Science**
GPA: 3.52
Physics (Core Pathway)
Stanford University

2020 – 2024 — **Associate of Science for Transfer**
SUMMA CUM LAUDE (GPA: 4.00)
Physics, Mathematics, Engineering, & Philosophy
Los Angeles City College

PUBLICATIONS & MANUSCRIPTS

Ocampo-Gómez, J. R., & Petrosian, V. (2026). Cosmological evolution of flat-spectrum radio quasars and the intrinsic gamma-ray–radio luminosity correlation. *In preparation for ApJ.*

Ocampo-Gómez, J. R., Yu, Z., & Allen, S. W. (2026). Fe II reverberation lags in OzDES active galactic nuclei. *In preparation.*

FELLOWSHIPS & AWARDS

2026 — **VPUE STEM Fellow**
Stanford University

2025 — **Physics SURP Funding**
Stanford University

2024 — **Jack Kent Cooke Transfer Scholarship**
Jack Kent Cooke Foundation

2023 — **Physics & Engineering Department Scholarship**
Los Angeles City College

2023 — **Mathematics Department Scholarship**
Los Angeles City College

2022 — **Eugene McKnight Philosophy Scholarship**
Los Angeles City College

POSTER PRESENTATIONS

Stanford Physics Summer Undergraduate Research Program (SURP) (2026). The cosmological evolution of gamma-ray and radio luminosity functions in FSRQs

Stanford Symposium of Undergraduate Research and Public Service (SURPS) (2026); **Stanford Physics SURP** (2025). Probing the Fe II emission region of AGNs with the OzDES RM project

National Conference for Undergraduate Research (NCUR) (2024); **Southern California Conference for Undergraduate Research (SCCUR)** (2023). Flight data telemetry device with data logging using customized off-the-shelf modules for experimental radio control aircraft

PUBLIC OUTREACH

SEPT. 2025 - PRESENT

KIPAC Public Outreach
Team Member

- Presented career-focused talks and led scientific demonstrations for underrepresented high school students
- Facilitated public stargazing nights at the Stanford Student Observatory, operating telescopes, managing attendee flow, and engaging visitors in discussions of astronomy
- Delivered astronomy talks in Spanish as part of the Noches Astronómicas outreach program, including moderated Q&A sessions

JUNE 2026 - PRESENT

Stanford Community College Outreach Program (CCOP)
Mentor

- Mentored community college students on the STEM transfer process, attending program workshops alongside them and leading follow-up discussions in small groups of eight to ten.
- Advised mentees on applications, coursework, and program selection, and connected them with resources specific to transferring into STEM fields.

LEADERSHIP POSITIONS

MARCH 2026 - PRESENT

Stanford SSI
Policy Team Co-Lead

- Leading a project to draft a model regulatory framework for amateur and high-power rocketry in Uruguay, an activity currently outside any dedicated national licensing or airspace regime.
- Assessing the case for Uruguay as a regional hub for amateur rocketry, addressing launch-site siting, third-party liability, and state responsibility obligations under Article VI of the Outer Space Treaty.
- Co-authored a policy paper on preventing the deployment of nuclear weapons in orbit; leading additional work on satellite traffic management.
- Managing small research teams and setting the agenda across international space law, space security, and nuclear nonproliferation.

JUNE 2026 - PRESENT

Stanford Transfer Network (STN)
Co-President

- Helped promote community among transfer students and alumni.
- In charge of finances and budgeting for social events, as well as communication and planning.

TALKS

Invited Research Talk - MESA Program, College of San Mateo Planetarium (2026). Listening to the Echoes of Supermassive Black Holes.

TRiO CenCal Conference, Evergreen Valley College - KIPAC Public Outreach (2026). Magic with Lights.

Noches Astronómicas - KIPAC Public Outreach (2025). Agujeros Negros: Los motores invisibles del Universo.

PROJECTS

Bayesian AGN SED Modeling (Stanford Physics 113, 2026)

- Developed a Python implementation of thin-disk spectral energy distribution fitting for AGN.
- Applied Metropolis–Hastings MCMC to infer black hole masses and accretion rates from observational data.
- Reproduced literature measurements for high-redshift blazars and investigated sensitivity to numerical integration methods.

COMPUTER SKILLS

Python/Jupyter, C++, Mathematica, LaTeX, UNIX, Git/GitHub, MicroPython, Java/Processing

COMMUNICATION SKILLS

Public Outreach, Scientific Communication, Bilingual Fluency (Spanish), Management/Supervising

STUDENT MEMBERSHIPS

American Physical Society (APS) (Since 2025)
National Society of Hispanic Physicists (NSHP) (Since 2025)
Society of Physics Students (SPS) (Since 2025)